



AI for Fuel Efficiency Optimization in Transportation Fleets

Case Study: Egyptian Truck Transportation Sector

STUDENT

Ramy Emel Ayoub

SUPERVISING PROFESSOR

Ahmed El Sedawy



The Egyptian Freight & Logistics Opportunity

EGP 420B

Logistics Sector Value

Core economic artery driving regional commerce.

800,000+

Commercial Vehicles

Largest active fleet footprint on the African continent.

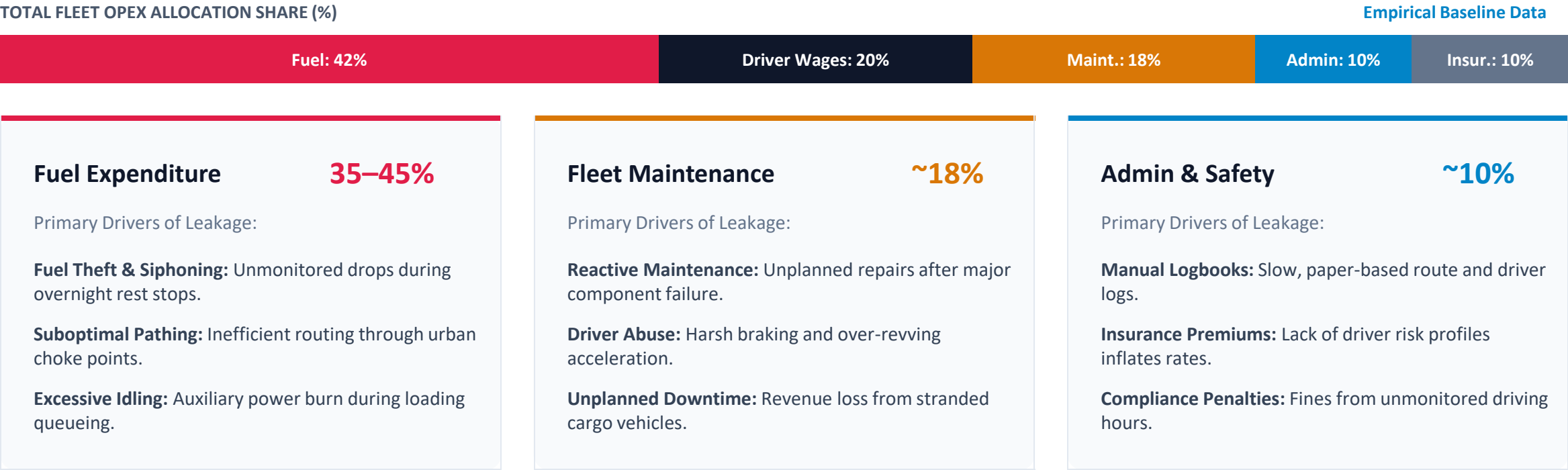


Digital Stagnation: < 18% Telematics

Heavy reliance on paper logs, unmonitored routes, and zero real-time operational visibility.



The Business Problem: Fleet OPEX Cost Breakdown



The Telematics Link: Capturing Physical Reality



1

Structured IoT Sensor Inputs

OBD-II/CAN-bus capture GPS coordinates, fuel levels, RPM, and engine fault codes.

2

Unstructured Computer Vision

Dual dashcams capture real-time driver fatigue, lane departures, and road hazards.

3

Cellular Cloud Streaming

Ultra-low latency MQTT/HTTP protocols beam telemetry into Azure/AWS cloud lakes.

Architecture of the Cognitive AI Engine



Highway Bottleneck Simulation: Ring Road Case Study

FREE FLOW SPEED

90 km/h

21 L / 100 km baseline

BOTTLENECK QUEUE

9 km/h

+167% Fuel Burn (24.4 L/100km)

ANNUAL ECONOMIC LOSS (700 VEHICLE SIMULATION)

EGP 38.5 M

Direct Annual Fuel Waste

EGP 78.3 M

Total Economic Loss (Fuel + Delay)

AI DYNAMIC REROUTING IMPACT

Optimization Results

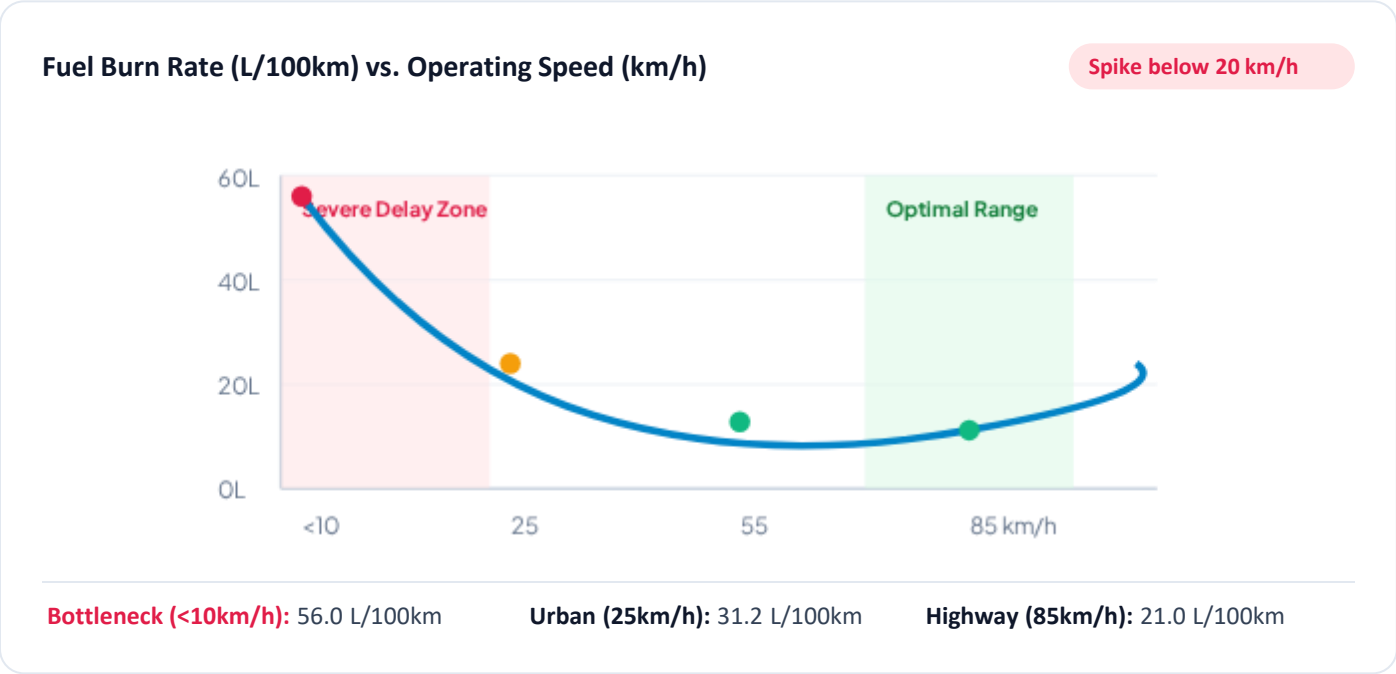
Fuel Saved / Trip: ~ 12%

Delay Reduction: ~ 18 Min / Trip

CO2 Emissions Reduced: ~ 8%

"Dijkstra-guided dynamic pathing proactively redirects traffic before vehicles enter bottleneck bottlenecks."

Speed vs. Fuel Burn: Non-Linear Efficiency Curve



⚡ Exponential Fuel Spike Below 20 km/h

Fuel consumption increases by **+166.6%** in heavy congestion due to continuous gear-shifts and idle friction.

🕒 22.4% Idle Fuel Waste Ratio

Over 1/5th of daily fuel expenditure generates zero vehicle mileage during loading docks and toll queueing.

🛣️ The Dynamic Rerouting Pivot

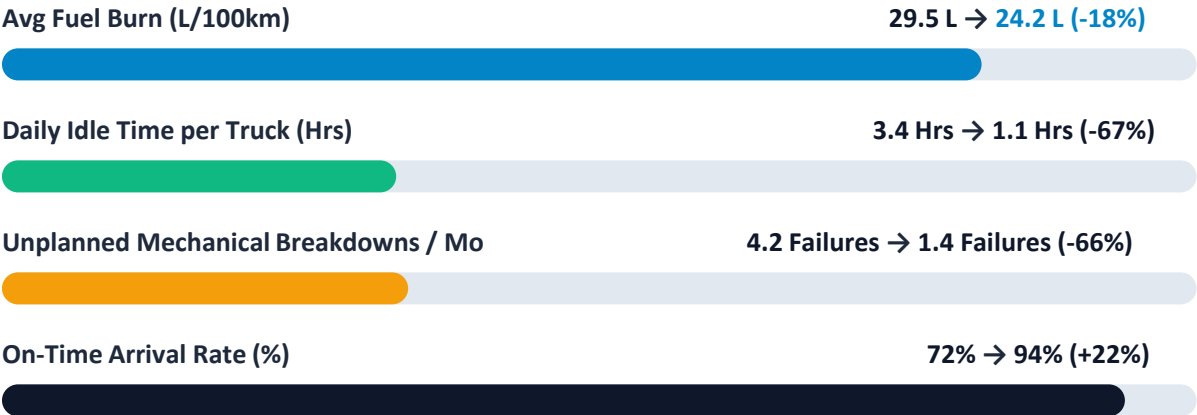
Rerouting around bottlenecks saves **12% fuel** even if distance increases by up to 8 km.

Operational Paradigm Shift: Analog vs. AI Era

Dimension	Traditional Analog Era	The AI-Driven Era
Fleet Visibility	< 18% Blind Tracking (Paper Logs)	100% Real-Time GPS & Telemetry
Maintenance Model	Reactive Repair (Fix after breakdown)	Predictive ML (Preventive alerts)
Workflow Dispatch	Manual calls & intuition-based pathing	Dynamic Rerouting via Dijkstra / RL
Driver Evaluation	Zero performance visibility	Computer Vision Safety & Risk Scores
Fleet Uptime	Unplanned breakdowns & delays	Continuous Uptime Optimization

Performance Comparison: Baseline vs. AI Fleet Standard

Key Operational Benchmarks



15% Net ROI

Direct OPEX Reduction

Achieved via real-time rerouting, idle caps, and predictive maintenance.

< 7 Mos

Hardware Payback

Rapid telematics ROI turnaround.

100%

Live Visibility

Full fleet command tracking.

Securing the Fleet Ecosystem (ISO 27001 ISMS)



Risk Assessment

Continuous threat mapping across edge IoT sensors and cloud data infrastructure.



Access Control

Strict RBAC permissions ensuring dispatchers only view authorized data layers.



Data Classification

GPS tracks, fuel metrics, and dashcam videos strictly encrypted and classified.

Supplier Security

Mandatory ISO compliance enforced on AWS/Azure cloud host platforms.



Incident Response

Documented rapid-recovery protocols for potential system or gateway breaches.



Continuous Audits

Regular penetration testing, security audits, and staff cybersecurity training.

Phased Enterprise Implementation Roadmap

PHASE 1 (MONTHS 0–3)

Pilot & Baseline

Hardware Rollout: Equip 50 pilot commercial vehicles with IoT gateways.

Data Collection: Gather baseline telematics and fuel burn profiles.

KPI Benchmarking: Establish ground truth metrics for route efficiency.

Milestone: Pilot Operations Active

PHASE 2 (MONTHS 3–6)

AI Engine Integration

Model Training: Train Random Forest and K-Means models on local data.

ERP Integration: Connect dispatch alerts directly to company ERP.

Driver Testing: Launch safety scoring and fatigue alerts.

Milestone: Predictive Modules Live

PHASE 3 (MONTHS 6–12)

Full Commercial Scale

Full Scale Fleet: Hardware installation across entire fleet.

Live Optimization: Continuous Dijkstra rerouting active.

Automated RPA: Continuous model refinement and reporting.

Milestone: Full ROI Realization



Transforming Egyptian Fleet Operations into a Sustainable Competitive Advantage

By combining IoT telematics with predictive machine learning models, commercial fleets can unlock **15% fuel cost savings**, prevent critical breakdowns, and protect operating margins against rising energy costs.

Thank You for Your Attention

Open for Academic Discussion & Board Defense

AASTMT MBA Project Defense

Image Sources



<https://vantage-logistics.com/wp-content/uploads/2026/08/vlc-news-vinfast-ha-tinh-29082026-2.png>

Source: vantage-logistics.com



<https://cdnimg.co/c1f03142-7cf5-485a-88c7-02db58ab6620/bb370001-27b1-4e97-997c-3ac9a4aed9aa/ai-dashcam-for-hgv-fleets-fleet-safety.jpg>

Source: fleetalyse.co.uk